Wanja Anastasia Tolksdorf Summer semester 2023

Dawn of a new religion? –

An analysis of dataism using Durkheim's sociology of religion

Contents

1. Introduction 3
2. Harari's definition of religion and dataism 4
3. Durkheim's definition of religion and his example of totemism 6
4. Analysis of dataism using Durkheim's characteristics of religion 8
   4.1. A religion is a unified system of beliefs and practices. 8
   4.2. Sacred things are given moral superiority and reverence. 9
   4.3. Sacred things are things set apart and forbidden. 10
   4.4. The sacred requires ritual reinforcement. 11
   4.5. The sacred unites into one single moral community called a Church. 12
   4.6. The sacred serves group interests, creates social solidarity and commitment to the group. 13
5. Conclusion 15
6. References 17

1. Introduction

In recent years a decline in traditional religiosity was witnessed in many world regions (Jackson et al., 2023). But following Durkheim, it is unlikely that all forms of religion will vanish. He argues that religion, or some substitute for it, will always exist due to the social function it provides (Durkheim, 1915, as cited in Pals, 2015). Parallel to this development, there has been a rise of big data and artificial intelligence along with predicted disruptions so big that leading artificial intelligence experts call for a pause in the development of artificial intelligence ('Pause Giant AI Experiments', 2023). Jackson et al. (2023) have shown that the decline of traditional religions is correlated with an increase in automation, that means robotics and artificial intelligence.

Harari (2017) also takes the view that new religions emerge in Silicon Valley rather than the Islamic state or the Bible belt. There are currently two main types of potential new religions: techno-humanism and dataism (Harari, 2017). According to Harari (2017), techno-humanism faces the following dilemma. Like humanism, it views the human experience and will as the source of all meaning and authority (Harari, 2017). It also wants to control the human will, that means to change and manipulate it (Harari, 2017). But once it has achieved that, it would not know how to deal with it. That is because, as soon as human will can be changed, it cannot function as the source of meaning anymore (Harari, 2017). Therefore, Harari (2017) makes the argument that the most interesting potential new religion is dataism. This alleged religion not only postulates that the universe consists of data flows, but also worships the data, or rather the data processing system, values freedom of information and wants to expand the data flow (Harari, 2017).

However, apart from Harari who has a pretty broad definition of religion, most scholars have avoided to conceptualize dataism as a religion (Uribe, 2023). By looking at dataism as religion this paper aims to address this research gap. Checking dataism for religious characteristics will help to illuminate dynamics that otherwise remain unnoticed. One of the only researchers who has started doing this work is Uribe (2023). He investigated if dataism can be included in the religious field using Bourdieu's sociology of religion and concluded that it has, in fact, a lot of religious traits (Uribe, 2023). I argue that Durkheim's conception of religion and his distinction between the sacred and the profane (Durkheim, 1915) can contribute another valuable perspective on dataism as religion.

So, to continue this line of research, I use Durkheim's sociology of religion to tackle the question proposed by Uribe (2023): *Does dataism have enough specific features to be included in the religious field?* First, Harari's definition of religion and dataism will be reviewed, followed by a summary of Durkheim's sociology of religion including his case study of totemism. Key characteristics of religion

will be extracted from Durkheim and used for the analysis of dataism. Throughout this analysis, social media will serve as an example of dataism to make the abstract terms more concrete.

2. Harari's definition of religion and dataism

Dataism is first of all a unifying scientific theory which emerged due to recent developments in the biosciences that state that organisms are just biochemical algorithms and concurrent advances in the creation of electronic algorithms in the computer sciences (Harari, 2017). Others include more disciplines and term it the "nano-bio-info-cogno convergence" (Latzer, 2021, p. 7). This convergence led to dataism which states that the same mathematical laws apply to biochemical and electronic algorithms (Harari, 2017). If we look at the world through the lens of dataism, the universe consists of data flows and everything is interpreted as data processing system: organisms and societies, economies and political systems (Harari, 2017). Capitalism and communism, democracy and dictatorship are competing data processing systems, distributed versus centralized ones (Harari, 2017). Likewise, humanity is a data processing system, humans are microchips and history is a process to increase the efficiency of the system (Harari, 2017). However, the question remains, why does Harari call dataism a religion?

To answer this question, we need to take a step back and look at Harari's definition of religion first. He defines religion as an all-encompassing story which provides human norms, values, social structures and laws with a superhuman legitimation (Harari, 2017). In other words, religion states that there are moral laws that humans did not create and cannot change, but need to obey (Harari, 2017). It thus tells us what is right and wrong or good and bad and what we ought to do. According to Harari (2018), the power of humanity is based on mass cooperation which requires the belief in common fictional stories. In his definition, a religion is nothing else than such a story. Religions thus provide social order and enable mass cooperation (Harari, 2017). Furthermore, they unite certain people while they differentiate them from their neighbours (Harari, 2018). Most successful religions have the additional characteristic of being missionary, meaning they attempt to spread to everyone (Harari, 2017). Lastly, religions make use of rituals. Harari (2018) defines a ritual as a magical act which makes the fictional real. And because suffering is the most real thing in the world, he believes making sacrifices is the most powerful ritual of all (Harari, 2018). Making someone suffer for a fictional story is enough to make them believe the story is true because people do not like to admit that they were fooled, and all their sacrifices were pointless (Harari, 2018).

An interesting aspect of Harari's view on religion is that he does not need a god in order to explain religion and the history of humanity as Behr (2020) notices. In other words he is a functionalist (Behr, 2020). Religion has the function of providing order and enabling cooperation of the masses. It also

becomes apparent that Harari's definition of religion is quite broad. He calls religion what others would call ideology and includes theistic religions like Islam and Christianity, as well as so-called humanistic religions like liberalism and communism (Harari, 2017; SelfAwarePatterns, 2020). While the former worship one or multiple gods, the latter worship humans and the human experience (Harari, 2017). According to Harari (2017), humanism is the current religion which might soon be replaced by dataism.

Now we can turn to the question what makes dataism a religion in Harari's view. Instead of worshiping gods or humans, dataism is a story that worships the data (Harari, 2017). Its supreme value is freedom of information, which grants the information the right to circulate freely and is not to be confused with freedom of speech which gives humans the right to say what they want, but also to keep quiet (Harari, 2017). Dataism also has a goal: to expand the data flow (Harari, 2017). This shows that dataism is a religion in the sense that it imposes moral laws and claims to determine what is right and wrong (Harari, 2017). To maximize the data flow is good, to block it resembles a sin (Harari, 2017). Even the value of each organism is measured according to its contribution to data processing (Harari, 2017). Dataism is also missionary: everyone and everything should be connected to the system, even those who do not want to (Harari, 2017). Only the question if and how dataism unites people, creates social order and cooperation remains largely unanswered by Harari. Because dataism gives people a common goal and tells them what to do, one can imagine that it creates a certain order, but Harari does not explicitly talk about this aspect.

Essential for the following analysis is what the core of the potential new religion is. In other words, what is sacred in dataism according to Harari? As mentioned before, the goal of dataism is to expand the data flow. That is the purpose of the current biochemical data processing system including humanity (Harari, 2017). More precisely, the goal is to create a more efficient data processing system and this new system is what Harari (2017) calls sacred (without defining the term "sacred"). Harari (2017) hints at the fact that the system will consist of Big Data and computer algorithms by pointing out that they are more efficient at data processing, but does not deliver more specific information. Latzer's (2021) description of digitalization seems fitting to further illuminate the nature of this electronic data processing system and how it is emerging. He views the digitalization we witness in the 21st century as a co-evolution of datafication, algorithmization and platformization (Latzer, 2021). Datafication refers to the production of big data by voluntary disclosure of personal data in exchange for online services (e.g. search terms), the collection of metadata (e.g. location) and the generation of data by objects equipped with sensors (also called the internet of things) (Latzer, 2021). One can imagine a lot of smart objects - from fridges to cars - many of which exist already today. So, data collection happens through digital devices (computers, phones, smart objects) and on the internet. To make use of this data, it is automatically processed by electronic algorithms (increasingly based on

artificial intelligence) which Latzer (2021) terms "algorithmization" (p. 5). Lastly, the "*platformization* of markets" (Latzer, 2021, p. 6) enables further datafication and algorithmization. Prominent examples include social media platforms like Facebook and Instagram, but platforms rise in all kinds of sectors: take for example Uber, a platform that replaces taxi drivers.

Sharing experiences on social media platforms like Facebook is also an example used by Harari (2017). He describes, if you travel to India and see an elephant there, you do not experience the moment, but take a picture and post it on Facebook (Harari, 2017). This is common practice today because dataism postulates that experiences are only valuable if they are shared and contribute to the data flow (Harari, 2017). In this example, the three parts of digitalization can be found, too: taking a photo of the elephant turns this experience into a chunk of data (datafication), then it is uploaded on the internet to the platform Facebook (platformization) and processed by algorithms which extract information about the user or determine whom to recommend the post (algorithmization). This digital data processing system is seen as sacred if we follow Harari's line of thought and will just be referred to as "data processing system" in the remainder of this paper. However, it remains to be analysed if the data processing system also fits Durkheim's definition of the sacred.

To facilitate and focus the analysis, the example of social media will be used to illustrate different aspects of dataism. It must be kept in mind though that "[d]igital technology is not limited to one specific technology" (p. 14) and that "it is not so much about assessments of specific technology, but of the underlying world views" (p. 14) as Latzer (2021) states. Care must be taken to not reduce dataism to the example of social media because it encompasses much more than that. Other examples can look quite different, although they too fall under dataism and show the same traits of datafication and algorithmization: say a smart watch used to track health and fitness. I will refer back to these other examples whenever the social media example with its specificities might bias the analysis of dataism too much. That being said, the next step towards an analysis of dataism using Durkheim's sociology of religion is to summarize his view on religion.

3. Durkheim's definition of religion and his example of totemism

In his book "The Elementary Forms of the Religious Life" (1915) Durkheim defines religion as follows:

> *A religion is a unified system of beliefs and practices relative to sacred things, that is to say, things set apart and forbidden—beliefs and practices which unite into one single moral community called a Church, all those who adhere to them*. (p. 47)

There are several things to unpack in this definition, starting from the notion of the sacred. Contrary to the conventional view, Durkheim (1915, as cited in Pals, 2015) does not believe the supernatural to

be the core element of religion, but the sacred. Religious people divide the world into two spheres, the sacred and the profane (Durkheim, 1915, as cited in Pals, 2015). Sacred things are superior and are treated with reverence (Durkheim, 1915, as cited in Pals, 2015). Furthermore, as the definition above states, sacred things are "*things set apart and forbidden*" (Durkheim, 1915, p. 47). Profane things on the other hand "belong to the ordinary, uneventful, and practical routine of everyday life" (Durkheim, 1915, as cited in Pals, 2015, p. 91). Important to note is that the sacred should not be equated with good and the profane with evil. Both can be good or evil (Durkheim, 1915, as cited in Pals, 2015). The sacred (as opposed to the profane) is what religion is concerned with.

In the second half of the definition Durkheim (1915) reveals that the function of the sacred is to "*unite into one single moral community called a Church, all those who adhere to them*" (p. 47). That means that sacred things are concerned with the interests and welfare of a whole group or community while profane things are about personal matters (Durkheim, 1915, as cited in Pals, 2015). This is also what distinguishes religion from magic which only satisfies private needs (Durkheim, 1915, as cited in Pals, 2015). The sacred instead has the function to create social solidarity and commitment to the group. Last, but not least, the sacred requires ritual reinforcement: During religious rituals "individuals renew their commitment to the community, reminding themselves in the most solemn fashion that they depend on the clan, just as it depends on them" (Durkheim, 1915, as cited in Pals, 2015, p. 100).

Kuhn (1982, as cited in Uribe, 2023) stated in order to really understand concepts, one needs to study an exemplary case instead of just looking at the theory. Therefore, Durkheim's analysis of totemism, an Australian tribal religion which he studied as a simple exemplary religion (Durkheim, 1915, as cited in Pals, 2015), will be shortly reviewed here. The clans of Australian tribal people have their own totem animal or plant which is sacred (Durkheim, 1915, as cited in Pals, 2015). Therefore, it is forbidden to kill and eat it like other (profane) animals (Durkheim, 1915, as cited in Pals, 2015). Totem beliefs also shape how the world is perceived, for example how natural objects are categorized and put in a hierarchy of power (Durkheim, 1915, as cited in Pals, 2015). But clans do not just worship the animal, they worship an impersonal force found within the animal called the totemic principle (Durkheim, 1915, as cited in Pals, 2015). What is more, "[t]he totem, in brief, is simultaneously the symbol of both the god *and* the clan, because both the god and the clan are in reality the same thing" (Durkheim, 1915, as cited in Pals, 2015, p. 96), so what is actually worshipped is the clan itself.

The essential ingredient of totemism, however, are the rituals or cults which are used to renew the members' commitment to the clan (Durkheim, 1915, as cited in Pals, 2015). The negative cult keeps the sacred apart from the profane by prohibitions, for example the prohibition to follow profane everyday activities on certain holy days (Durkheim, 1915, as cited in Pals, 2015). It often purposely imposes pain on the individual to teach people to sacrifice their own needs in favour of group interests

(Durkheim, 1915, as cited in Pals, 2015). The positive cult on the other hand provides rituals to move the clan into the realm of the sacred at selected occasions (Durkheim, 1915, as cited in Pals, 2015). Only then is it allowed to eat the totem animal (Durkheim, 1915, as cited in Pals, 2015). Durkheim (1915, as cited in Pals, 2015) explains that these rituals represent a sacred exchange in which people receive something in return for their loyalty to the clan. It also shows that the clan depends on the individual and the individual depends on the clan (Durkheim, 1915, as cited in Pals, 2015). On top of that, there are piacular rites which follow a person's death or catastrophe and provide the clan with new strength after losing a member (Durkheim, 1915, as cited in Pals, 2015).

This example shows the importance Durkheim ascribes to the social function of religion and how rituals are used to reinforce the commitment to the group again and again. The functionalist view on religion and relevance given to rituals is something that Durkheim and Harari seem to have in common. There are, however, also considerable differences in their definitions of religion. While Harari focuses on the prescription of values, goals and actions, Durkheim (1915, as cited in Pals, 2015) sees beliefs as secondary and rituals as more fundamental. He also weighs in on the importance of the sacred and its service to the community. Harari does not seem to ascribe enough importance to the term "sacred" to even define it and although he talks about religions creating social order, he does not come back to this aspect during his explanation of the new religion dataism. Therefore, it remains interesting, how Durkheim's sociology of religion transfers to dataism.

To structure the analysis of dataism, the following key characteristics shall summarize Durkheim's definition of religion and the sacred:

1) A religion is a unified system of beliefs and practices relative to sacred things which…
2) …are given moral superiority and reverence.
3) …are things set apart and forbidden.
4) …require ritual reinforcement.
5) …unite into one single moral community called a Church.
6) …serve group interests, create social solidarity and commitment to the group.

4. Analysis of dataism using Durkheim's characteristics of religion

4.1. A religion is a unified system of beliefs and practices.

To determine if dataism can be included in the religious field according to Durkheim's sociology of religion, it must first be assessed if it has a unified system of beliefs and practices. As described above, there are some strong beliefs at the core of dataism: the belief that the data processing system is sacred, that freedom of information is the ultimate value and that the data flow should be expanded

(Harari, 2017). Just like totemic beliefs shape tribal people's world view, these beliefs shape how dataists – the adherents of dataism – see the world and what they prioritize: the organism, economic system or technology that is most efficient at data processing is seen as superior. Based on these values, dataism also prescribes certain practices revolving around maximizing the data flow. Everyone should produce, share and consume as much data as possible (Harari, 2017).

This is easily demonstrated using the example of social media. Dataism requires you to capture and share every elephant - but also more routine activities like the meal you had today – for the whole world to see on Facebook, Instagram or similar platforms. Harari (2017) even suggests that this behaviour is a question of survival: by posting we prove that we are still valuable. To summarize, this first question can be answered with a clear yes, there is a system of beliefs and practices at the core of dataism.

4.2. Sacred things are given moral superiority and reverence.

Next, it needs to be analysed if the data processing system which these beliefs and practices are related to is sacred in the Durkheimian sense. One characteristic of sacred things is that they are given moral superiority and treated with great respect or reverence. Alexander (1990) recognizes that the conversion of any message into a series of numbers by a computer seems like a rationalization and disenchantment of the world. But his review of the discourse on computers in the middle and second half of the 20th century shows that the computer was seen as a sacred, mysterious object with godly powers (Alexander, 1990). Instead of rational scientific reporting, a mythical discourse surrounded the computer, either promising salvation or predicting an apocalypse (Alexander, 1990). In the meantime, we have, of course, gotten used to computers and they have lost some of their mysteries.

But Campolo & Crawford (2020) have recently made a similar observation about artificial intelligence:

> On one hand, it represents a complex form of technological calculation and prediction, phenomena Weber associated with disenchantment. On the other hand, both deep learning experts and observers deploy enchanted, magical discourses to describe these systems' uninterpretable mechanisms and counter-intuitive behavior. (p. 1)

This combination of a magical discourse and predictive accuracy that characterizes artificial intelligence is what Campolo & Crawford (2020) coin "*enchanted determinism*" (p. 3).

Furthermore, Tsuria (2021) notes that the internet, too, has been conceptualized as sacred, divine or magical by a number of scholars. Even the terms used for digital technology like "The Cloud, Big Data, and Data Lake" (Tsuria, 2021, p. 6) sound mythical and convey power. Latzer (2021) summarizes the previous arguments perfectly and points out the resulting blind trust in technology:

> As digital technology, increasingly perceived as superior, takes on a god-like function in the form of the internet, computers and artificial intelligence, there has been a resurgence of faith as blind trust, but now as faith (i.e. without knowledge) in digital technology (for example in rationally incomprehensible big-data predictions of automated algorithmic-selection applications), which is increasingly becoming the basis for action and thus for decisions. (p. 21)

In the case of social media sites, it is often astonishing how well the algorithmically chosen recommendations fit the user's current situation and desires. It goes to the extent that it seems like magic or mind-reading. A particularly extreme case are "if you see this" videos on TikTok (Cotter et al., 2022). If they are shown on the user's algorithmically curated home page, they convey the message that this video was meant for them (Cotter et al., 2022). Cotter et al. (2022) use the term "algorithmic conspirituality" (p. 3) to describe "spiritualizing beliefs about algorithms, which emerge from occasions when people find personal, often revelatory connections to content algorithmically recommended to them" (p. 3). People with these beliefs essentially see such events as "algorithmically mediated cosmic intervention" (Cotter et al., 2022, p. 3).

This section can be concluded with the result that the data processing system is in fact seen as superior and inspires awe. Interestingly, dataism is based on the belief that everything is calculable and still, many parts involved in the data processing system (the computer that serves as the hardware, the internet used for data sharing and artificial intelligence algorithms which automatically process the data) are perceived as magical. Tsuria (2021) deduces, "if technology is indistinguishable from magic, and magic is synonymous with religion, then technology becomes indistinguishable from religion" (p. 5). That magic is synonymous with religion is, however, an erroneous assumption, since Durkheim emphasises that there is a major difference between the two: magic satisfies private needs, religion serves group interests (Durkheim, 1915, as cited in Pals, 2015). Therefore, it cannot yet be concluded that dataism belongs to the religious field, not before it has been tested against the remaining characteristics of religions.

4.3. Sacred things are things set apart and forbidden.

One of those characteristics is, that sacred things are things set apart and forbidden. Alexander (1990) has already demonstrated that this was true for the early computers in the 20th century. They were secluded in special buildings inaccessible to the public (Alexander, 1990). Only experts had access to them and knew how to use them (Alexander, 1990). Some of these observations still hold true today. The massive computers used to keep up the internet, train artificial intelligence algorithms or store the vast amounts of data generated each day, are still kept in data centres belonging to large companies or research institutes and are thus physically isolated from the general public (e.g. *About Google Data Centers*, n.d.). On top of that, it is more difficult than ever to develop and understand

these algorithms and the amount of resources, in particular data and computing power, required for the training of large algorithms and processing of big data reserves the power over the technology for selected few (West & Vipra, 2023). On the other hand, since the invention of the personal computer more and more people have gained access to it. Likewise, the internet has become accessible to an increasing number of people since its advent (Tsuria, 2021). Now, anyone with an internet connection can read as well as post, that means anyone has access to the power that was once possessed by few (Tsuria, 2021). This is especially true for social media platforms where you can create a profile and post stuff within seconds. Through those platforms everyone uses not only the internet, but also artificial intelligence algorithms every day at least indirectly because they curate the posts on the starting page for example. But recently a more explicit communication with artificial intelligence by lay people has been made possible, too, for example by the release of the large language model ChatGPT to the public.

To summarize, most of the data processing system is not set apart from profane everyday life, neither in space nor in time. Dataism certainly does not forbid its usage like totemism does with the totem animal. Instead, "it is used by 'everyone' […] for almost anything" (Tsuria, 2021, pp. 7–8) and that is what dataism encourages.

4.4. The sacred requires ritual reinforcement.

What needs to be examined next, are the rituals connected to dataism. Sen (2017) states that the

> "informational milieu" is also a breeding ground for ritual. First, activities in the digital domain take on a regular and repetitive aspect that is central to all ritual. Digital acts such as searching, liking, tagging, or retweeting are automatic and sometimes even automated (as when performed by social bots). Second, as I hope to argue in the next section, the consequences of these acts also resemble those of ritual in that they consist of external impositions on the individual that he or she has no control over. (p. 477)

Thus, interactions with the data processing system like sharing information are highly ritualistic, at least in terms of being habitual, repetitive, and imposed. It is easy to find this repetitive behaviour, but much harder to find examples that resemble the different cults that Durkheim distinguishes in his interpretation of totemism, especially the cult that keeps the sacred apart from the profane. The reason for this difficulty lies in the fact that, in the case of dataism, the sacred is not clearly set apart from the profane, as was established in the previous chapter. The (allegedly sacred) data processing system is rather seamlessly integrated into everyday life and often used for profane activities, too (Tsuria, 2021). Consequentially, there are also no rites that set it apart. Or the other way around, the sacred is not set apart from the profane because of a lack of rituals providing that function. Most digital

rituals seem to fit best into the positive cult which moves people into the realm of the sacred. They reinforce the connection of users to the data processing system and act as a reminder that the users depend on the system as much as the system depends on them.

How such rituals sustain engagement and solidarity was investigated by Burroughs (2014) who conducted a case study: he used the example of Facebook and FarmVille to analyse the rituals of social media, more specifically social gaming. He found different kinds of rituals, for example exchange rituals where digital labour is exchanged to advance in the game (Burroughs, 2014). These rituals lead to reciprocity and feelings of closeness (Burroughs, 2014) which points to the next characteristic of religions.

4.5. The sacred unites into one single moral community called a Church.

The sacred unites people into one single moral community called a Church. This seems to be the case for social media as Burroughs (2014) describes:

> Members of social networks enacting ritual scripts through mediated technology, who then feel a sense of belonging or social cohesion through that behavioral action, enact a *real* sense of belonging that operates like a myth to make the collective seem real. (p. 155)

Furthermore, Burroughs (2014) suggests that social networks are so-called digital third places, environments that support the creation of emotional bonds among people which is something that traditionally takes place in bars and pubs. Social media, however, creates many different communities, say a community of travellers, one of FarmVille players, one of artists and many more. These would be multiple Churches or rather chapels as Durkheim would say, and not one single moral community. Moreover, other examples of dataism do not necessarily create such communities. A fitness tracker for example belongs to the data processing system because it is used to collect and share data. As long as it does not have community built into the app and is just used to privately track the users' fitness (and feed the data to algorithms), it does not create a community like social media does. But is there one single community that unites all dataists, regardless which app they use or which kind of data they contribute to the data processing system?

At this point it might be helpful to take a closer look at Durkheim's definition of a moral community or a Church:

> A society whose members are united by the fact that they think in the same way in regard to the sacred world and its relations with the profane world, and by the fact that they translate these common ideas into common practices, is what is called a Church. (Durkheim, 1915, p. 44)

If a Church is just defined by the fact that its members have the same beliefs and practices, then there is a Church of dataism. As was outlined earlier, there are common beliefs and practices that all dataists follow. But reading a bit further, Durkheim (1915) points out that believers in the same religion create lasting bonds to their god and between each other. In contrary, the magician has clients that do not form lasting relations with the magician or the other clients (Durkheim, 1915). What is the case for dataism?

It can be argued that there is an ongoing bond to the data processing system. Social media for example is not something that is used once or twice when needed, but rather people continue scrolling and posting regularly. These ritualised practices ensure a continuous engagement. Sometimes people might switch from one platform to another, but that does not diminish the fact that they are still engaging with social media or the data processing system as a whole. Most people cannot imagine being cut off from the data processing system. "For the approximately 2.5 billion users of the Internet in the world today, the idea of a self that is solely offline is simply not tenable" argues Sen (2017, p. 478). No wonder this is the case: people would lose their meaning and value in the dataist belief system if they stopped participating in data generation and processing (Harari, 2017). This establishes a strong dependence on and lasting bond to the data processing system.

Relations between the believers are more difficult to find. As Burroughs (2014) has shown, there are different sub-groups, especially on social media, that form close, lasting bonds. But at the same time there are people who use, for example, above-mentioned fitness tracker. The data of all these people might be aggregated in one place, but they never get to know each other. Between the fitness tracker users – and especially between adherents of dataism in general – there is not necessarily a lasting connection.

To put it into a nutshell, the answer to the question if there is a Church of dataism is ambivalent. Although dataists believe and act the same, establish a long-lasting relationship with the data processing system and form connected sub-groups, they do not necessarily form a united group with lasting relations to the other members.

4.6. The sacred serves group interests, creates social solidarity and commitment to the group.

Continuing the thought of the previous chapter, what also distinguishes religion from magic is that the latter is used to satisfy profane personal needs while the former serves group interests, creates social solidarity and commitment to the group. Burroughs (2014) has shown that social media and social gaming does create reciprocal relationships and solidarity with the group. On the other hand, there are enough examples in which the data processing system is only used to satisfy personal needs. Tsuria

(2021) argues that the internet – which plays an essential role in the data collection and processing system – is mainly used for ordinary everyday tasks nowadays, for example for work, shopping or entertainment purposes. This usage for private matters makes the internet profane (Tsuria, 2021). The same can be said for the fitness tracker example from the previous chapter.

However, people's use of digital services only represents the surface of the data processing system. At its core it is about the collection and processing of the data traces that people leave while using digital devices, browsing the internet and posting on social media. Since users disclose their personal information, they sacrifice their privacy (Dijck, 2014) like tribal people sacrifice their own interests for the sake of the group. But is the sacrifice of privacy a sacrifice for the sake of the group or for other reasons? Dataist missionaries point to the advantages of the freedom of information and processing of big data (Harari, 2017). For example, Google claims to detect new epidemics quicker than it is detected on traditional ways (Harari, 2017). If you are convinced by these arguments and therefore decide to share your personal information, it is a clear sign of commitment to the group. But is this the case for the average digital technology user?

There is an alternative explanation of the sacrifice which probably resonates with a lot more users. Dijck (2014) says "users have to give up part of their privacy in exchange for free convenient platform services. In other words, meta data appear to have become a regular currency for citizens to pay for their communication services and security" (p. 197-198). This sounds much more like a profane deal between a magician and his clients to describe it with Durkheim's terminology. After data collection the personal information is sold to advertisers or data companies (Dijck, 2014). Researchers, companies and government agencies alike have an interest in extracting as much value as possible from the data (Dijck, 2014; Latzer, 2021). In essence, people's experiences are first turned into data, to be then turned into profits, for example through targeted advertisements. The sacred characteristics of parts of the data processing system come in handy for those who pursue the very profane goal of making money off people's participation in the system. The social is commodified (Latzer, 2021) and used for personal interests which makes it profane. In Tsuria's words, "[o]nce the moneychangers entered the Temple of the Web, it became desecrated" (Tsuria, 2021, p. 7).

For now, it can be concluded that it is questionable to what extent the data processing system serves group interests, while it is evident that it serves the personal interests of selected few. Solidarity is created within smaller groups of the data processing system, but the whole system is likely to be driven by barter instead of real solidarity.

5. Conclusion

Having investigated how six characteristics of religion apply to dataism, it is time to draw a conclusion. Is the data processing system sacred and can dataism be included in the religious field? The answer is not clear-cut. On one hand, it was found that dataism provides a unified system of beliefs and practices and the data processing system is sacred in some regards: it seems magical and superior and is treated with reverence. There are digital rituals which reinforce the user's engagement with the data processing system, so that a long-lasting relation to the data processing system is formed. On the other hand, the data processing system is not clearly set apart from the profane. Connections and solidarity are created within small sub-groups, but there is no single united community of dataists. The data processing system seems to serve personal interests more than group interests and instead of solidarity, the convenience of free services is likely to be the driver for the continuous use of the data processing system. This shows that the data processing system has sacred as well as profane characteristics.

Trying to resolve this tension of sacred and profane, Tsuria (2021) argues that the technology itself cannot be sacred because it is used for profane activities. Instead, we have sanctified humans who attained god-like powers through technology (Tsuria, 2021). Humans and their mundane desires, experiences and activities are sacred (Tsuria, 2021). This is in line with Harari (2017) who also thinks that this so-called humanism is the current dominant religion. Tsuria (2021) continues to elaborate that the internet only becomes sacred because it represents those sacred human experiences. This theory can also be applied to datafication more generally which essentially doubles the world based on digital data (Latzer, 2021). So, for Tsuria humanism stays the dominant religion. But she acknowledges that data sharing is the new moral obligation (Tsuria, 2021). This contradicts her theory since there seems to be a considerable difference between valuing the human experience in itself and valuing the conversion of human experience into data and the sharing of this data. The former are humanistic, the latter dataist beliefs and Tsuria's theory can, at the very most, explain the shift from humanism to dataism.

On one hand, it cannot be denied that something new is emerging. On the other hand, one has to agree with Tsuria in that the data processing system is sacred and profane at the same time. Therefore, dataism cannot be included in the religious field, yet. Conspicuous is that dataism fulfils all the criteria that magic and religion have in common. "Magic, too, is made up of beliefs and rites" (Durkheim, 1915, p. 42) and beliefs and rites, as well as magical awe-inspiring properties, are the most pronounced features of dataism. Meanwhile, the relation of dataism to criteria that distinguish religion and magic, namely that religion unites people into a Church, serves group interests and creates solidarity with the group are criteria where the position of dataism is at least ambiguous. Therefore, it suggests itself that

dataism falls in the category of magic rather than religion. This does not mean that dataism will forever belong into the field of magic. Since dataism is only starting to emerge right now, it is possible that it still develops into a religion in the future.

This is one of the limitations this research faces: it can only represent a snapshot of dataism as religion (or rather magic) right now and cannot speak for the future. Another hurdle was the broad nature of the concept of dataism. Although the focus on social media helped to make it more concrete, social media alone is an insufficient representation of the whole data processing system which posed further challenges in the analysis. Furthermore, an attempt was made to stay very close to Durkheim's sociology of religion and his original definitions. But scholars have begun to criticise aspects of his work and adapted his concepts to analyse contemporary issues of media and technology (e.g. Couldry, 2005). Theses adaptations could make Durkheim's theory more applicable to modern issues, so future research could adopt them to see if they are more suitable for an analysis of dataism. Another recommendation for future research is to address the limitations of this paper by looking at multiple other examples next to social media during the discussion of dataism and, of course, by contributing an updated view on dataism as religion a few years into the future.

6. References

About Google Data Centers. (n.d.). Google. Retrieved 14 September 2023, from https://www.google.com/about/datacenters/

Alexander, J. (1990). The Sacred and Profane Information Machine: Discourse about the Computer as Ideology. *Archives de Sciences Sociales Des Religions*, *35*(69), 161–171.

Behr, A. (2020). *Religionskritik bei Yuval Noah Harari*. RPI Loccum. Retrieved 3 September 2023, from http://www.rpi-loccum.de/material/pelikan/pel3_20/3-20_Behr_02

Burroughs, B. (2014). Facebook and FarmVille: A Digital Ritual Analysis of Social Gaming. *Games and Culture*, *9*(3), 151–166. https://doi.org/10.1177/1555412014535663

Campolo, A., & Crawford, K. (2020). Enchanted Determinism: Power without Responsibility in Artificial Intelligence. *Engaging Science, Technology, and Society*, *6*, 1–19. https://doi.org/10.17351/ests2020.277

Cotter, K., Decook, J. R., Kanthawala, S., & Foyle, K. (2022). In FYP We Trust: The Divine Force of Algorithmic Conspirituality. *International Journal of Communication*, *16*, 1–23.

Couldry, N. (2005). Media Rituals: Beyond Functionalism. In E. Rothenbuhler & M. Coman, *Media Anthropology* (pp. 59–69). SAGE Publications, Inc. https://doi.org/10.4135/9781452233819.n6

Dijck, J. van. (2014). Datafication, dataism and dataveillance: Big Data between scientific paradigm and ideology. *Surveillance & Society*, *12*(2), 197–208. https://doi.org/10.24908/ss.v12i2.4776

Durkheim, É. (1915). *The Elementary Forms of the Religious Life*. George Allen & Unwin Ltd. https://www.gutenberg.org/files/41360/41360-h/41360-h.htm

Harari, Y. N. (2017). *Homo Deus*. C.H.BECK. https://doi.org/10.17104/9783406704024

Harari, Y. N. (2018). *21 Lektionen für das 21. Jahrhundert*. C.H.BECK. https://doi.org/10.17104/9783406727795

Jackson, J. C., Yam, K. C., Tang, P. M., Sibley, C. G., & Waytz, A. (2023). Exposure to automation explains religious declines. *Proceedings of the National Academy of Sciences*, *120*(34). https://doi.org/10.1073/pnas.2304748120

Latzer, M. (2021). *Digital Trinity – Controllable Evolution – Everyday Religion. Characteristics of the Socio-Technical Transformation of Digitalization* [Working Paper – Media Change & Innovation Division]. https://doi.org/10.2139/ssrn.3854485

Pals, D. L. (2015). Society as sacred: Émile Durkheim. In D. L. Pals, *Nine theories of religion* (Third edition, pp. 81–111). Oxford University Press.

Pause Giant AI Experiments: An Open Letter. (2023, March 22). *Future of Life Institute*. Retrieved 13 September 2023, from https://futureoflife.org/open-letter/pause-giant-ai-experiments/

SelfAwarePatterns. (2020). The superhuman order definition of religion. *SelfAwarePatterns*. Retrieved 3 September 2023, from https://selfawarepatterns.com/2020/07/20/the-superhuman-order-definition-of-religion/

Sen, B. (2017). Information as Ritual: James Carey in the Digital Age. *Cultural Studies ↔ Critical Methodologies*, *17*(6), 473–481. https://doi.org/10.1177/1532708615625687

Tsuria, R. (2021). Digital Media: When God Becomes Everybody—The Blurring of Sacred and Profane. *Religions*, *12*(2), Article 2. https://doi.org/10.3390/rel12020110

Uribe, D. A. (2023). A new religion for the 21st century: An interpretation of 'dataism' from the sociology of pierre bourdieu. *Вестник КазНУ. Серия Психологии и Социологии*, *85*(2), Article 2. https://doi.org/10.26577/JPsS.2023.v85.i2.012

West, S. M., & Vipra, J. (2023, June 22). Computational Power and AI. *AI Now Institute*. Retrieved 14 September 2023, from https://ainowinstitute.org/publication/policy/computational-power-and-ai